Background subtraction for electron diffraction frames: what it is, the mathematics, how far it reaches

Subproject **pod/bgsub** (GitHub mit-eremin/pod-subtract), Dmitry Eremin, Nelson Lab, Caltech, 2026-11-07. Package **bgsub** (Python + numba, 47 tests). Goal in the owner's words: a fast, mathematically correct, universal background ESTIMATION AND SUBTRACTION for diffraction frames, better than the flat local boxes of XDS / DIALS; background only, not an integrator, not for XDS. Page 1: what the algorithm does and why it is correct; page 2: how it looks on real frames from three detectors and where the project stands.

1. What is going on

A diffraction frame is Bragg spots on a smooth, bright, structured background (beam halo, film rings, stop shadow and penumbra, grid bars). Integration programs estimate it locally and flatly (a box around each spot) and lose intensity on a curved background; screening programs threshold a band-passed image and never estimate it. bgsub fits **one smooth model of the whole frame's background in the detector's own stored counts**, with weights and outlier tests from the detector's declared count statistics, and returns the background map, its per-pixel variance and the subtracted frame. Versions v0-v2.1 (v1 default; v2.1 adds re-centring and an elliptical ring refinement), all frozen and reproducible.

2. The mathematics

$$K_p \sim \text{law}\,(\mu_p), \quad \mathrm{E}[K_p] = b_p = B(r_p, \varphi_p), \quad \mathrm{Var}[K_p] = g(b_p)$$

Observation model. K_p is the stored value of pixel p. Its mean is a smooth background B at the pixel's polar position; its variance is a known function g of the mean, the detector's **count law**: g(b) = b for a Poisson counter; for the Apollo the law of round-half-even(3-frame electron sum / m) with tie rule and dead time (sub-linear at bin 8, 16); g(b) = gain·b + read variance for an integrating camera. Only g is declared; no distribution shape is assumed.

$$B(r, \varphi) = \sum_k \theta_k \phi_k(r, \varphi)$$

Linear smooth model. A fine cubic B-spline in radius (rings, the steep halo) times azimuthal harmonics m ≤ 6 on a coarse radial spline, plus four derivative columns that absorb a centre error and an ellipticity, plus penumbra columns on the 1-4 pixels next to the beam stop. Everything is linear in the coefficients θ; the frame is first reduced to sufficient statistics on polar cells (1-px rings x 64 sectors x 2 checkerboard halves): cell sums S_c and counts n_c.

$$\hat{\theta}_A = \arg\min_\theta \sum_{c \in A} w_c\,(S_c - n_c X_c \theta)^2, \qquad w_c = \frac{1}{n_c\, g(\hat{b}_c^{(B)})}$$

Cross-fitted weighted least squares. Half A of the checkerboard is fitted with weights computed from the OTHER half's estimate of the mean, and vice versa; the two fits are averaged. Because the weights do not depend on the data being fitted, the estimator stays **linear in the counts**: E[the estimate of θ] = θ exactly whenever the true background lies in the model's span, **for any count law**: the law affects only the efficiency (weights, thresholds, variances), never the bias. No median, no Poisson assumption, no iteration on the same data.

$$P(K_p > k^*(\hat{b}_p)) = 10^{-6}, \qquad P\Big(\sum_{3\times 3} K > k_9^*\Big) = 10^{-6}$$

What is kept out. Spots, hot pixels and zingers are excluded by one-sided tail tests of the declared law, per pixel and per 3 x 3 window, dilated by one pixel; spots the caller will measure are always excluded by discs. Excluded pixels never enter the fit, so the background under a spot is what the surrounding smooth model says, not the spot's own tails (the flat-box failure).

$$\mathrm{Var}[\hat{\theta}] = \hat{\phi}\,(X^\top W X)^{-1} X^\top W \Sigma W X (X^\top W X)^{-1}, \quad \hat{B}_p = X_p \hat{\theta}, \quad \mathrm{Var}[\hat{B}_p] = X_p \mathrm{Var}[\hat{\theta}] X_p^\top$$

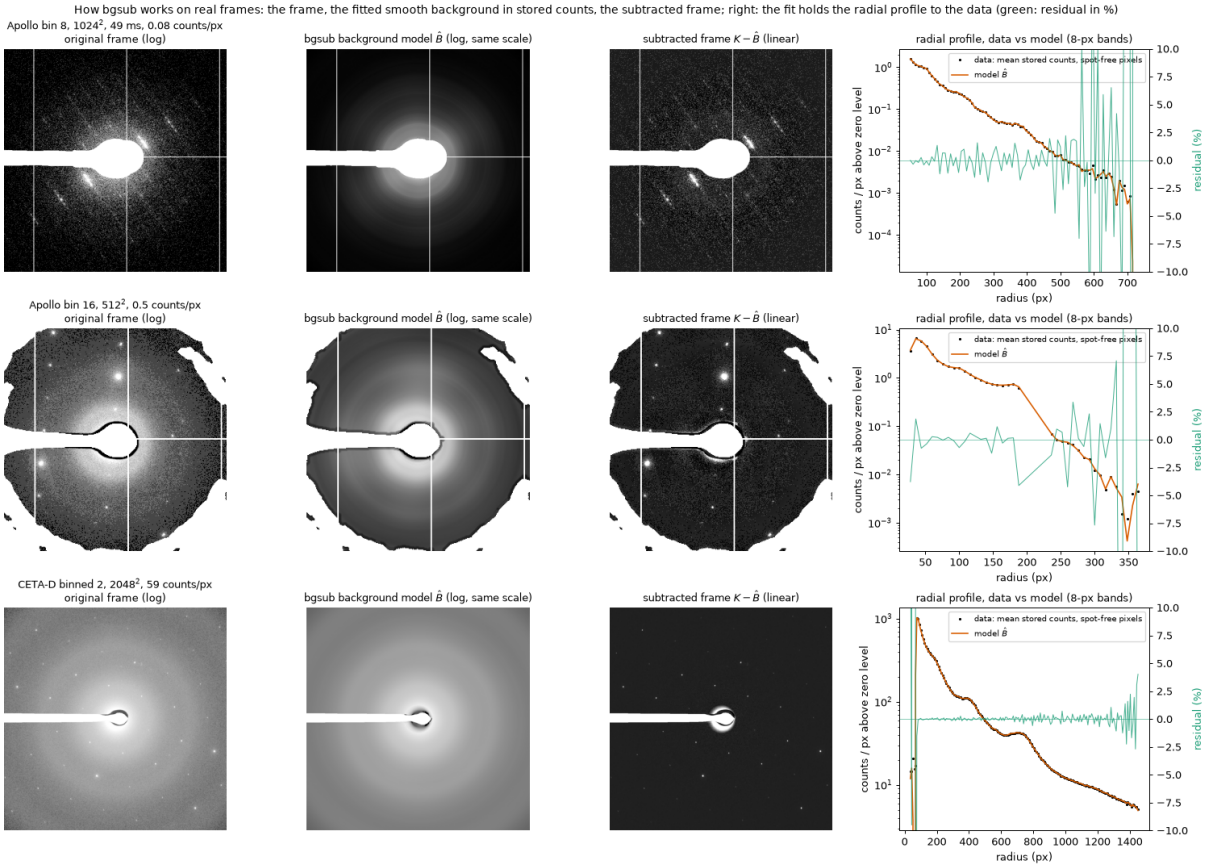
Honest variance. A sandwich covariance with the frame's own Pearson dispersion ϕ■ (the ratio of observed to law variance on the used pixels) gives a per-pixel variance of the background map; the subtracted frame K - B carries g(B) + Var[B]. The dispersion is itself a diagnostic: 1.02 on Apollo bin 8 real frames, i.e. the law and the fit agree with the data.

Measured on real Apollo frames (compare v0, 2026-11-05): the fitted background matches the spot-free data to within 1 % in every 32-px radial band at bin 4 and bin 8 (bin 8: mean and variance agree with the law to 10^{-4} counts / px); a flat local annulus on the same spots removes 15 % more 'background' and loses 5-14 % of the net intensity (17 % on bright spots); the XDS table x one scalar is 9-14 % low in the halo. Fit time 60-160 ms per 1024^2-2048^2 frame.

3. How universal it is

- **Universal by construction:** the unbiasedness argument above holds for any detector whose stored value has a declared mean-variance law, any geometry the centre / ellipticity columns and the mask can express, any stop (penumbra columns from the mask), any binning. Nothing is tuned per sample.
- **Declared per detector, not inferred:** the count law g(b) and the zero level. Apollo: the stored-count law per binning (m = 1, 4, 16 e per count, tie rule, dead time, TIFF pedestal) is measured and built in; tested on real bin 4, 8 and 16 frames. CETA-D and TVIPS: a Poisson(gain, read noise) law exists but is NOT calibrated, and compare v1 showed their noise is spatially correlated (a 3 x 3 window carries ~3 x the variance of nine independent pixels), so the law must carry a correlation term; with a declared estimate both already run (real binned-2 movies). Not yet: Apollo bin 2 (4096^2).
- **Not universal:** structure outside the model's span is not modelled: a film ring sharper than the spline knots (one classic session), the 20-40 px glow at the edge of a CETA-D / TVIPS stop, a sample's own diffuse scattering at the annulus scale. These are the known limits, each with a named anomaly.

4. What it looks like on real frames



Three detectors / binnings from compare v1 (2026-11-05): Apollo bin 8 fly-scan snapshot at 0.08 counts / px (top), Apollo bin 16 grossular at 0.5 counts / px (middle), CETA-D grossular rotation frame at 59 counts / px (bottom). Columns: the original frame (log), bgsub's fitted background B on the same scale, the subtracted frame K - B (linear: a flat field with the Bragg spots standing alone), and the radial profile of spot-free pixels against the model with the residual in percent (green). The CETA-D row uses the declared read-noise + correlation law estimated from the frame; the glow crescent right of its stop is the known unmodelled edge effect. Side-by-side panels against pyREyes and XDS / superxds for all five datasets: notes/reports/2026-11-05_compare_v1_panels.pdf.

5. Where we are

	status (2026-11-07)
Algorithm	v0-v2.1 implemented, frozen, 47 tests; v1 default, v2.1 opt-in (re-centring, elliptical ring refinement, gated local layer). Fit 60-160 ms per frame, plan cached.
Correctness, synthetic (truth known)	60 lacey cells x bin 4 / 8 x dose x structure: global estimator unbiased in 60 / 60 cells; the hybrid fails 1 cell at 0.2 counts per box where the test had no statistical power. Count statistics / truncation: pass (24 cells).
Correctness, real Apollo	Background matches spot-free data to < 1 % in every radial band (bin 4, 8); better than a flat local background in every injected-peak regime (flat loses 5-14 % intensity). Peaks per frame bgsub / pyREyes / XDS: bin 4 156 / 50 / 34, bin 8 22 / 20 / 4, bin 16 53 / 19 / 2.
Yardsticks	pyREyes 3.6.3 re-implemented exactly (no background model); superxds = XDS ported to numpy and validated against XDS (tables 99.98 % exact, spot lists identical): a sweep-averaged table x one scalar, 9-14 % off in the halo, fails on single sparse bin-8 frames.
Other detectors	CETA-D and TVIPS run with a declared law estimated from the frame (gain 3.5 / 10, read variance 8 / 0, correlation factor 2.8 / 3.1); peaks 101 / 20 / 12 and 34 / 9 / 12. A calibrated law from dark / flat frames is the next real-data task.
Verdict by its own pre-registered rules	v2.1 NOT VALID by two narrow margins (one sparse synthetic cell; 54 vs 50 ms); the thresholds were the orchestrator's, and the owner ruled timing relative. A background-only validity spec with owner-endorsed thresholds is ROADMAP item 1.

6. Next steps and the owner's decisions

Next: 1. a background-only validity spec, pre-registered (bias everywhere and under masked spots with stated power, variance honesty, stop / bar / crystal zones, count-law regimes; timing reported); 2. v2.1 to valid under it; 3. a calibrated integrating-detector law with noise correlation, then CETA-D / TVIPS validation on the movies held locally; 4-5. Apollo bin 2, bin-16 variance. **Owner decisions:** (a) thresholds to endorse; (b) per-frame output format (proposed: float32 background + variance maps in stored counts, subtracted frame on request, sidecar); (c) CETA-D / TVIPS sessions and dark / flat frames to calibrate on; (d) the XDS-merge mov11 beam origin is 75 px off, never refined by XDS: re-check.